

# Uniform Random Mapping Functional-Graph Closure: Exact Cycle–Component Counts and Marked-Orbit Limits

Route-A source-local certificate HCS-C276

1 September 2026

## Abstract

For a uniform random function  $f : [n] \rightarrow [n]$ , we give a self-contained uniform random mapping functional-graph closure. Decomposing the graph into a permutation and a rooted forest yields the exact joint count of cyclic vertices and components, its marginal, and every expected cycle count. An ordered-prefix argument gives the complete tail–cycle law of a marked orbit, identifies its collision length in distribution with the number of cyclic vertices, and proves their common Rayleigh limit. Conditional uniformity then gives the two-dimensional limiting density  $e^{-(x+y)^2/2}$ ; exhaustive independent computation audits every map through  $n = 7$  without replacing the all-size proof.

## 1 Model and theorem

Choose each of the  $n^n$  functions  $f : [n] \rightarrow [n]$  with equal probability. Its directed functional graph has edges  $v \rightarrow f(v)$ . Let  $C_n$  be the number of cyclic vertices and  $K_n$  its number of weak components. Write  $c(k, r)$  for the unsigned Stirling number of the first kind and  $(n)_k = n(n-1)\cdots(n-k+1)$ .

**Theorem 1** (Finite and asymptotic closure). *For  $1 \leq r \leq k \leq n$ ,*

$$\#\{f : C_n = k, K_n = r\} = \binom{n}{k} c(k, r) k n^{n-k-1}, \quad (1)$$

where the last factor means the unique empty forest when  $k = n$ . Hence

$$\mathbb{P}(C_n = k) = \frac{(n)_k k}{n^{k+1}}, \quad \mathbb{E} Z_{n,\ell} = \frac{(n)_\ell}{\ell n^\ell}. \quad (2)$$

where  $Z_{n,\ell}$  counts directed  $\ell$ -cycles. Fix a starting vertex. If  $\mu$  is its tail length,  $\lambda$  its eventual cycle length, and  $R_n = \mu + \lambda$ , then

$$\mathbb{P}(\mu = u, \lambda = \ell) = \frac{(n-1)_{u+\ell-1}}{n^{u+\ell}} \quad (u \geq 0, \ell \geq 1, u + \ell \leq n). \quad (3)$$

$R_n \stackrel{d}{=} C_n$ , and both variables divided by  $\sqrt{n}$  converge to the Rayleigh density  $x e^{-x^2/2} \mathbf{1}_{x \geq 0}$ . Moreover,

$$(\mu/\sqrt{n}, \lambda/\sqrt{n}) \Rightarrow (U, V), \quad f_{U,V}(x, y) = e^{-(x+y)^2/2} \mathbf{1}_{x,y \geq 0}. \quad (4)$$

These distributions are classical random-mapping statistics [1, 2]. Our contribution is a source-locked proof synthesis and executable boundary audit, not a literature-priority claim.

## 2 Permutation and forest proof

Fix the set  $S$  of  $k$  cyclic labels. The restriction to  $S$  is a permutation, and  $c(k, r)$  permutations have  $r$  cycles. Every label outside  $S$  lies in an in-tree directed toward a root in  $S$ . The corresponding complete-graph Laplacian minor, of size  $m = n - k$ , is

$$L_S = nI_m - J_m.$$

For  $m > 0$  its eigenvalues are  $k$  once and  $n$  with multiplicity  $m - 1$ , so  $\det L_S = k n^{n-k-1}$ . At  $m = 0$  the empty determinant is one. Choosing  $S$ , its permutation, and this forest is bijective with the maps in (1).

Summing over  $r$  and using  $\sum_r c(k, r) = k!$  gives

$$\#\{f : C_n = k\} = \binom{n}{k} k! k n^{n-k-1}.$$

After division by  $n^n$ , this is the first formula in (2). There are  $\binom{n}{\ell}(\ell-1)! = (n)_\ell/\ell$  possible directed  $\ell$ -cycles. Each prescribes  $\ell$  independent function values and has probability  $n^{-\ell}$ , so indicator linearity proves the second formula. In particular,  $n = 1$  gives the single fixed-point map, while  $k = n$  reduces (1) to  $c(n, r)$  as it must for permutations.

### 3 Marked orbit and the Rayleigh bridge

Put  $t = u + \ell$ . Before its first repeat, the marked orbit visits  $t$  distinct labels. After the fixed starting label, the ordered prefix has  $(n-1)_{t-1}$  choices. Its next edge is forced to the prefix label at index  $u$ , and all  $n-t$  unexposed function values remain arbitrary. The cell therefore contains  $(n-1)_{t-1}n^{n-t}$  maps, proving (3).

For each  $t$  there are exactly  $t$  possible tail positions. Consequently

$$\mathbb{P}(R_n = t) = \frac{t(n-1)_{t-1}}{n^t} = \frac{(n)_t t}{n^{t+1}} = \mathbb{P}(C_n = t). \quad (5)$$

which proves the finite distributional identity. It also proves that  $\mu$  conditional on  $R_n = t$  is uniform on  $\{0, \dots, t-1\}$ .

No collision during the first  $m$  transitions has probability

$$\mathbb{P}(R_n > m) = \frac{(n-1)_m}{n^m} = \prod_{j=1}^m \left(1 - \frac{j}{n}\right). \quad (6)$$

For  $m = \lfloor x\sqrt{n} \rfloor$ , logarithmic expansion gives  $\log \mathbb{P}(R_n > m) = -m(m+1)/(2n) + O(m^3/n^2) \rightarrow -x^2/2$ . The bound  $\log(1-z) \leq -z$  supplies tail tightness, so (6) converges to the Rayleigh survival function. Equation (5) transfers the limit to  $C_n$ . The boundary cells  $\mu = 0$ ,  $\lambda = 1$ , and  $t = n$  are already included in (3), without limiting conventions.

### 4 The two-dimensional limit

Let  $W_n = \mu/R_n$ . Given  $R_n = t$ , this variable is uniform on the grid  $\{0, 1/t, \dots, (t-1)/t\}$ . The Rayleigh limit implies  $R_n \rightarrow \infty$  in probability. Conditional Riemann sums therefore show

$$(R_n/\sqrt{n}, W_n) \Rightarrow (S, W),$$

where  $S$  has density  $se^{-s^2/2}$  and  $W$  is independent uniform on  $[0, 1]$ . Under  $(s, w) \mapsto (sw, s(1-w))$ , the Jacobian is  $s$ ; cancellation with the radial factor gives (4). Its normalization is transparent:

$$\int_0^\infty \int_0^\infty e^{-(x+y)^2/2} dx dy = \int_0^\infty se^{-s^2/2} ds = 1.$$

### 5 Executable certificate and scope

Exact receipts enumerate all 873,612 maps for  $1 \leq n \leq 7$ , closing 84 cycle-component cells, 84 marked cells, and 28 cycle-length aggregates. Formula atlases add 528 cyclic masses, 560 collision tails, and 528 cycle expectations; 28 high-precision samples audit the scaling code. A producer-independent orbit tracer closes 821 assertions, SymPy closes 918 identities, fresh replay is byte-identical, and repaired-hash hostile testing rejects 24/24 changes. These computations audit formulas and boundaries; Theorem 1 is proved above.

Under `NO_BAD_EULER_OR_ROOT_NUMBER`, ensemble averages supply no arithmetic local datum, Euler factor, root number, automorphy statement, target divisor, functional equation, or Hilbert–Pólya operator. There is also no natural same-clock self-adjoint operator for this random ensemble. The tuple is

$$(\text{AO\_FAIL}, \text{A1\_FAIL}, \text{A2\_FAIL}, \text{A3\_FAIL}, \text{A4\_FAIL}),$$

the verdict is `ROUTE_A_REJECTED`, and Route B is disabled.

### References

- [1] B. Harris, *Probability distributions related to random mappings*, Ann. Math. Statist. **31** (1960), 1045–1062, doi:10.1214/aoms/1177705677.
- [2] P. Flajolet and A. M. Odlyzko, *Random mapping statistics*, Advances in Cryptology—EUROCRYPT ’89, LNCS 434 (1990), 329–354, doi:10.1007/3-540-46885-4\_34.